

OUC Smart Beamer Template User Guide and Reference

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Template guide for XeLaTeX workflow

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What Is Included

This folder contains a reusable Beamer template extracted from the current presentation, with the visual rules moved into a standalone style package.

默认中文字体现已改为完整覆盖的 Songti SC，以保证所有常用简体中文字符都能正确编译；如需楷体，请使用 `\textkai{中文}`。

Core files

`oucsmartbeamer.sty`: theme, colors, title page, footer, boxes

`template-manual.tex`: this guide document

`fonts/`: bundled template fonts

`badge/`: bundled OUC and Adelaide badges

Use the style file for formatting, and keep your own deck focused on content only.

Folder Layout

```
1 beamertemplate/  
2 |-- oucsmartbeamer.sty  
3 |-- template-manual.tex  
4 |-- fonts/  
5 |-- badge/  
6 |   |-- AUlightbadge.png  
7 |   |-- ouc.png  
8
```

The default asset paths are now local to the template folder. If you rename or replace the bundled badges, update the corresponding path in \OUCBeamerSetup.

Available Commands (1/2)

Command	Purpose
<code>\OUCTitleFrame</code>	Draws the custom title slide
<code>\OUCClosingFrame</code>	Draws the custom closing slide
<code>\OUCBeamerSetup{...}</code>	Adjusts marker text, logos, footer text, closing text
<code>\OUCSetTitleFontSize{30}{34}</code>	Sets the cover title font size and line height
<code>\OUCSetBodyFontSize{\small}</code>	Sets body text size for normal content boxes and slide text
<code>\OUCSetHeaderFontSize{\Large}</code>	Sets the frame title size in the page header
<code>\OUCSetFooterFontSize{\footnotesize}</code>	Sets the footer text size
<code>\OUCSetPalette{N}</code>	Selects palette 0..23 (brand triple + box backgrounds)
<code>oucpal1 .. oucpal6</code>	Six numbered colors exposed by the active palette

Available Commands (2/2)

Command	Purpose
<code>\textkai{中文}</code>	Uses the bundled Kai style for selected Chinese text
<code>infobox</code> , <code>warnbox</code> , <code>resultbox</code> , <code>goalbox</code>	Reusable content boxes
<code>The</code> , <code>Exa</code> , <code>Rmk</code> , <code>Pro</code> , <code>Le</code> , <code>Co</code> , <code>De</code>	Book-style theorem-like environments
<code>Proof</code> , <code>Boxed</code>	Proof environment with QED, rounded grey container
<code>Intro</code> , <code>RightIntro</code>	Chapter-introduction boxes (left / right title)
<code>\UnderlineBox</code> , <code>\Quote</code>	Underlined block, formatted quotation
<code>\Minipage</code> , <code>\EvOdd</code>	Side-by-side layout, wrap-figure helper
<code>\Eq</code> , <code>\EqL</code>	Inline align wrappers (unlabelled / labelled)

Font Size Controls

```
1 \OUCSetTitleFontSize{30}{34}
2 \OUCSetBodyFontSize{\normalsize}
3 \OUCSetHeaderFontSize{\large}
4 \OUCSetFooterFontSize{\small}
5
```

The title size command takes two numbers: font size and baseline skip. The other three commands take standard LaTeX size switches such as `\small`, `\normalsize`, or `\Large`.

Recommended starting point: title 30/34, body `\normalsize`, header `\large`, footer `\small`.

LaTeX Font Size Switches

Sample text	<code>\tiny</code>	1	<code>{\tiny Sample text}</code>
Sample text	<code>\scriptsize</code>	2	<code>{\scriptsize Sample text}</code>
Sample text	<code>\footnotesize</code>	3	<code>{\footnotesize Sample text}</code>
Sample text	<code>\small</code>	4	<code>{\small Sample text}</code>
Sample text	<code>\normalsize</code>	5	<code>{\normalsize Sample text}</code>
Sample text	<code>\large</code>	6	<code>{\large Sample text}</code>
Sample text	<code>\Large</code>	7	<code>{\Large Sample text}</code>
Sample text	<code>\LARGE</code>	8	<code>{\LARGE Sample text}</code>
Sample text	<code>\huge</code>	9	<code>{\huge Sample text}</code>
Sample text	<code>\Huge</code>	10	<code>{\Huge Sample text}</code>
Sample text		11	

```
{\tiny Sample text}
{\scriptsize Sample text}
{\footnotesize Sample text}
{\small Sample text}
{\normalsize Sample text}
{\large Sample text}
{\Large Sample text}
{\LARGE Sample text}
{\huge Sample text}
{\Huge Sample text}
```

Palette Selection

Call `\OUCSetPalette{N}` once in the preamble. Indices 0–23 map to twenty-four predefined palettes; the active name is exposed via `\OUCPaletteName` (current: OUC Default).

```
1 \usepackage{oucsmartbeamer}
2 \OUCSetPalette{0} %% 0 OUC Default, 1 Brunneophobia, 2 Van Dyke,
3                 %% 3 Back in Black, 4 Belle of the Ball,
4                 %% 5 Pine Tree, 6 Provence Blue,
5                 %% 7 Fresco Blue, 8 Monet, 9 Narcissus,
6                 %% 10 Roman Empire, 11 Greece, 12 Kanagawa,
7                 %% 13 Starry Night, 14 A Thousand Li,
8                 %% 15 And Quiet Flows the Don,
9                 %% 16 Cyberpunk Edgerunners,
10                %% 17 The Grand Budapest Hotel,
11                %% 18 Renaissance Florence, 19 Soviet Avant-Garde,
12                %% 20 Constantinople, 21 France, 22 Kyoto,
13                %% 23 Siamese Dream
14
```

Configuration Keys

```
1 \OUCBeamerSetup{
2   marker=PROJECT REVIEW,
3   left logo=badge/AUlightbadge.png,
4   right logo=badge/ouc.png,
5   footer title=My Deck,
6   footer subtitle=Section Name,
7   closing title=Questions?,
8   closing subtitle=My Deck,
9   closing note=Thanks for listening
10 }
11
```

Standard Beamer metadata such as `\title`, `\subtitle`, `\author`, `\institute`, and `\date` remain the primary content inputs.

Example Chinese text: 这是一个默认中文示例，可直接用于课题汇报、课程作业和模板说明文档。局部楷体示例：這是楷體文字。

Box Environment Usage

```
1 \begin{infobox}
2   Background, definitions, and context.
3 \end{infobox}
4
5 \begin{warnbox}
6   Caveats, assumptions, and risks.
7 \end{warnbox}
8
9 \begin{resultbox}
10  Findings, conclusions, and key outputs.
11 \end{resultbox}
12
13 \begin{goalbox}
14   A short highlighted message.
15 \end{goalbox}
16
```

Built-in Environment Preview

Info box

Use for background, definitions, and explanations.

Result box

Use for outcomes, findings, and final numbers.

Warning box

Use for caveats, assumptions, or risks.

Goal boxes work well for short highlighted messages.

Listing Style Usage

```
1 \begin{frame}[fragile]
2   \frametitle{Code Example}
3   \begin{lstlisting}[language=Matlab]
4   for idx = 1:3
5     disp(idx)
6   end
7   %% the inner lstlisting and the frame are closed here
8
```

Code slides must use the [fragile] option. The template already applies the bundled matlabmodern listing style.

Listing Style Preview

```
1 for idx = 1:3
2     fprintf('Slide %d\n', idx);
3 end
4
```

Compile Command

```
1 cd beamertemplate
2 latexmk -xelatex -interaction=nonstopmode -halt-on-error template-manual.tex
3
```

Use XeLaTeX so that the bundled fonts and local badge assets are loaded correctly inside the template folder.

Definition 1: Group

A set with an associative operation, identity, and inverses.

Theorem 2: Lagrange

If $H \leq G$, then $|H|$ divides $|G|$.

```
1 \begin{De}{Group}{de:g}  
2   A set with an associative  
3   operation, identity, inverses.  
4 \end{De}  
5 \begin{The}{Lagrange}{the:lag}  
6   If  $H \leq G$ , then  $|H|$   
7   divides  $|G|$ .  
8 \end{The}  
9
```

Example 3

$\mathbb{Z}/6\mathbb{Z}$ is cyclic of order 6.

Remark 4

The trivial group $\{e\}$ is the unique group of order 1.

```
1 \begin{Exa}{}{exa:cyc}  
2    $\mathbb{Z}/6\mathbb{Z}$  is  
3   cyclic of order 6.  
4 \end{Exa}  
5 \begin{Rmk}{}{rmk:tr}  
6   The trivial group  $\{e\}$  is  
7   the unique group of order 1.  
8 \end{Rmk}
```

Environment: Proposition / Lemma / Corollary

Proposition 5: Closure

$H \cap K$ is closed under the group operation.

Lemma 6: Cancellation

If $ab = ac$ in a group, then $b = c$.

Corollary 7: Order divides

$|\langle g \rangle|$ divides $|G|$.

```
1 \begin{Pro}{Closure}{pro:cl}
2   $H\cap K$ is closed.
3 \end{Pro}
4 \begin{Le}{Cancellation}{le:can}
5   If $ab=ac$, then $b=c$.
6 \end{Le}
7 \begin{Co}{Order divides}{co:ord}
8   $|\langle g \rangle|$
9   divides $|G|$.
10 \end{Co}
11
```

Proof $x \in H \cap K \Rightarrow x \in H$ and $x \in K$; intersection is a subgroup. $\square$

Rounded grey container suited for paragraph-sized notes or summaries.

```
1 \begin{Proof}
2    $x \in H \cap K \Rightarrow x \in H$  and  $x \in K$ .
3 \end{Proof}
4 \begin{Boxed}
5   Rounded grey container for
6   paragraph-sized notes.
7 \end{Boxed}
8
9
```

Chapter Intro

Title attached to the upper-left corner; useful as a chapter or section opener.

Right Intro

Mirrored variant with the title on the upper-right corner.

```
1 \begin{Intro}{Chapter Intro}{Title  
   in the  
   upper-left corner.}  
2  
3 \end{Intro}  
4 \begin{RightIntro}{Right Intro}  
5   {Title in the upper-right corner.}  
6 \end{RightIntro}  
7
```

Underlined block with a double rule.

*Mathematics is the queen of the
sciences.*

C. F. Gauss

```
1 \UnderlineBox{Underlined block  
2   with a double rule.}  
3 \Quote{Mathematics is the queen  
4   of the sciences.}{C.\,F.\~Gauss}  
5
```

Left.
Side-by-side helper.

$$a^2 + b^2 = c^2$$

$$e^{i\pi} + 1 = 0$$

Right.
Use widths summing < 1.

(1)

```
1 \Minipage{0.46}{Left content}
2   {0.46}{Right content}
3 \Eq{a^2 + b^2 = c^2}
4 \EqL{e^{i\pi} + 1 = 0}{eq:euler}
5 %% \EvOdd{lines}{width}{content}
6 %%   = right-aligned wrapfigure
7
```

Background, definitions, and context.

Caveats, assumptions, and risks.

```
1 \begin{infobox}
2   Background, definitions,
3   and context.
4 \end{infobox}
5 \begin{warnbox}
6   Caveats, assumptions, risks.
7 \end{warnbox}
8
```

Environment: resultbox / goalbox

Findings, conclusions, and key outputs.

A short highlighted message.

```
1 \begin{resultbox}
2   Findings, conclusions,
3   and key outputs.
4 \end{resultbox}
5 \begin{goalbox}
6   A short highlighted message.
7 \end{goalbox}
8
```

Inline Math and Additional Packages

mathtools / amsmath

$$\frac{N}{D} \quad \binom{N}{D}$$

$$\frac{\text{above}}{\text{below}} \rightarrow$$

$$\overbrace{a+b}^c$$

mathrsfs

$$\mathcal{A}, \mathcal{B}, \mathcal{F}, \mathcal{L}$$

dsfont

$$\mathbb{R}, \mathbb{N}, \mathbb{Z}, \mathbb{Q}$$

polynomial

$$1 - x^2$$

`\tfrac`, `\tbinom`

`\xrightarrow[b]{a}`

`\overbracket`

`\mathscr`

`\mathds`

`\polynomial{...}`

nicefrac

$$3/4, a/b$$

accents

$$\underline{x}$$

$$\overset{*}{y}$$

dirtytalk

“Hello world”

extpfeil

$$A \xrightarrow{f} B$$

$$A \xrightarrow{g} B$$

`\nicefrac{...}{...}`

`\underaccent`

`\accentset`

`\say{...}`

`\xtwoheadrightarrow`

`\xhookrightarrow`

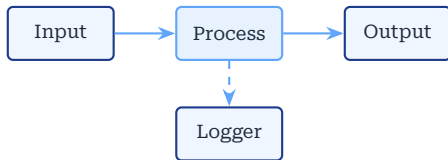
TikZ: Palette-Aware Drawing



```
1 \begin{tikzpicture}[scale=0.75]
2   \fill[paleblue]
3     (0,0) rectangle (4,1.4);
4   \draw[primary,thick,rounded
5     corners=2pt]
6     (0,0) rectangle (4,1.4);
7   \node[fill=primary,text=white,
8     rounded corners=2pt,inner
9     sep=3pt]
10    at (2,0.7) {Palette aware};
11  \draw[accent,very thick,-Stealth]
12    (0,-0.6) -- (4,-0.6);
13  \fill[oucpal5] (1.4,-1.8) circle
14    (0.25);
15 \end{tikzpicture}
```

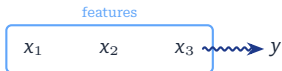
Tip: use palette-aware names `primary`, `accent`, `paleblue`, `oucpal1`–`oucpal6` for auto-recoloring. Pre-loaded: `positioning`, `arrows.meta`, `fit`, `calc`, `decorations.pathmorphing`, `shadows.blur`.

TikZ: Node Diagrams & Arrows



```
1 \begin{tikzpicture}[
2   node distance=6mm and 8mm,
3
4   box/.style={draw=primary,fill=paleblue,
5     thick,rounded corners=2pt,
6     minimum width=14mm,align=center},
7   hi/.style
8     ={{draw=accent,fill=accent!15,
9       thick,rounded corners=2pt,
10      minimum width=14mm,align=center},
11   arr/.style={-Stealth,accent,thick}]
12 \node[box] (a) {Input};
13 \node[hi, right=of a] (b)
14   {Process};
15 \node[box, right=of b] (c) {Output};
16 \node[box, below=of b] (d) {Logger};
17 \draw[arr] (a)--(b); \draw[arr]
18   (b)--(c);
19 \draw[arr,dashed] (b)--(d);
20 \end{tikzpicture}
```

TikZ: Highlights & Annotations



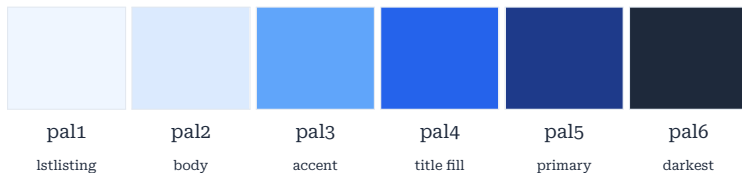
Result: $y = f(\mathbf{x})$

```
1 \begin{tikzpicture}
2   \node (n1) at (0,0)   {$x_1$};
3   \node (n2) at (1.0,0) {$x_2$};
4   \node (n3) at (2.0,0) {$x_3$};
5   \node (n4) at (3.2,0) {$y$};
6   \node[draw=accent,thick,
7     rounded corners=2pt,
8     fit=(n1)(n2)(n3),inner sep=4pt,
9     label=above:features] {};
10
11   \draw[primary,thick,-Stealth,decorate,
12     decoration={snake,amplitude=.6pt,
13       segment length=3pt}]
14     (n3.east) -- (n4.west);
15   \node[draw=primary,fill=paleblue,
16     rounded corners=2pt,
17     blur shadow] at (1.5,-1.2)
18     {Result:  $y=f(\mathbf{x})$ };
19 \end{tikzpicture}
```

All 24 palettes are reproduced on the following pages, each as its full pal1–pal6 swatch strip with the slot each colour drives. Switch to any of them with a single `\OUCSetPalette{N}` in the preamble.

Indices 0–9 are the original OUC set; 10–23 are the themed palettes shared with the Thomas-Tufte-Style-Book-Template, so a deck and its companion book can be set in exactly the same colours.

Palette 0 – OUC Default



OUC Default — deep navy blue. OUC house palette.

`\OUCSetPalette{0}`

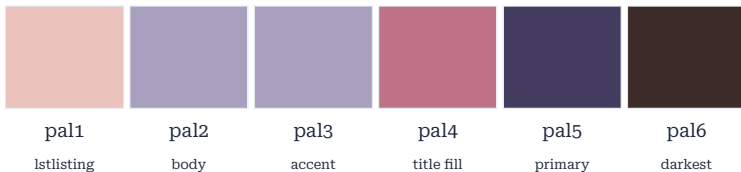
Palette 1 – Brunneophobia



Brunneophobia — warm earth tones, amber and deep brown.

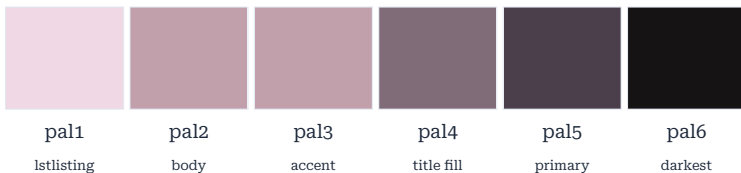
```
\OUCSetPalette{1}
```

Palette 2 – Van Dyke



Van Dyke — muted mauve and slate purple with warm rose mid-tones. `\OUCSetPalette{2}`

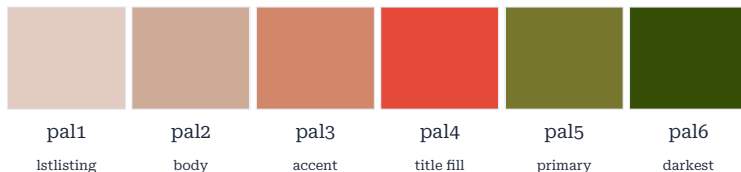
Palette 3 – Back in Black



Back in Black — dusty rose fading to near-monochrome charcoal.

```
\OUCSetPalette{3}
```

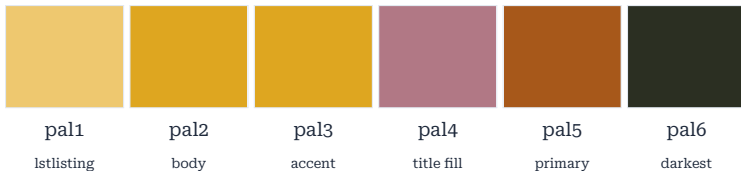
Palette 4 – Belle of the Ball



Belle of the Ball — terracotta orange and olive green, bold contrast.

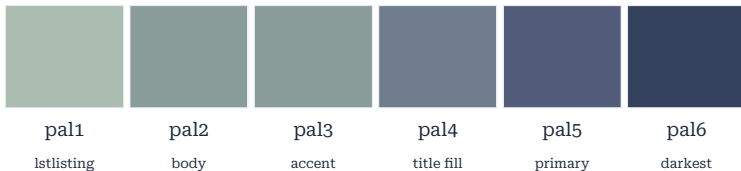
```
\OUCSetPalette{4}
```

Palette 5 – Pine Tree



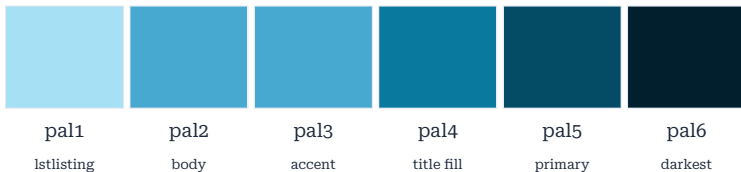
Pine Tree — golden amber with muted rose accents and deep forest brown.
`\OUCSetPalette{5}`

Palette 6 – Provence Blue



Provence Blue — sage green and slate blue, subdued Mediterranean tones.
`\OUCSetPalette{6}`

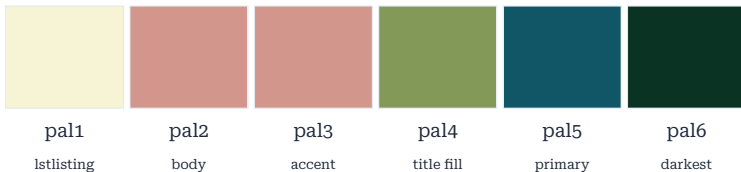
Palette 7 – Fresco Blue



Fresco Blue — bright sky blue grading into deep ocean teal.

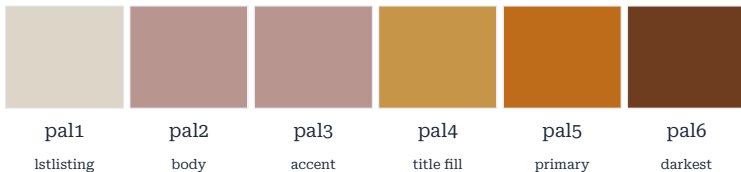
```
\OUCSetPalette{7}
```

Palette 8 – Monet



Monet — soft ivory, dusty coral and forest green, Impressionist palette. `\OUCSetPalette{8}`

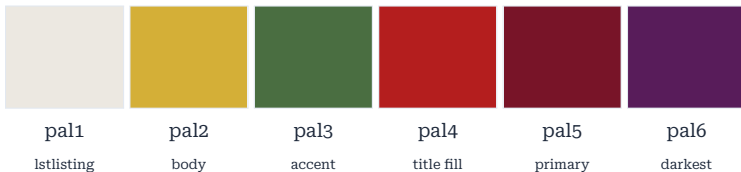
Palette 9 – Narcissus



Narcissus — sandy warm tones with a bold burnt-orange primary.

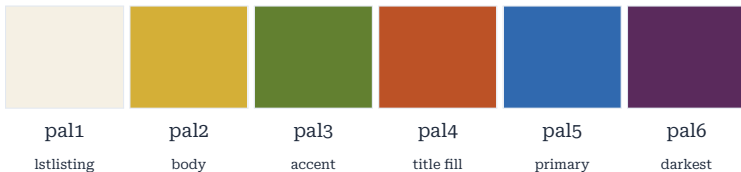
`\OUCSetPalette{9}`

Palette 10 – Roman Empire



Roman Empire — legion red and murex purple over imperial marble. `\OUCSetPalette{10}`

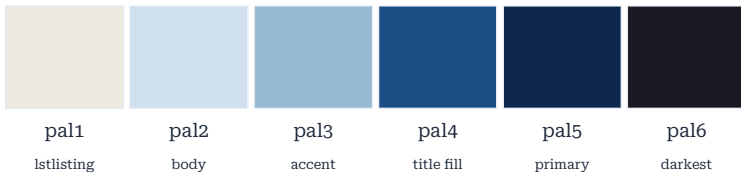
Palette 11 – Greece



Greece — Aegean blue, olive and terracotta on limestone.

`\OUCSetPalette{11}`

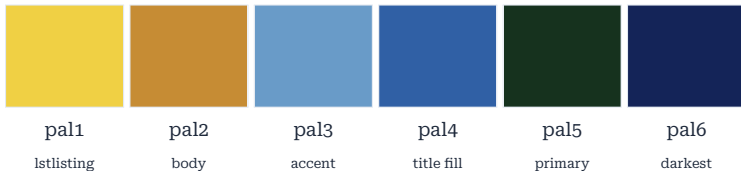
Palette 12 – Kanagawa



Kanagawa — Edo wave blues graded onto ink-black line work.

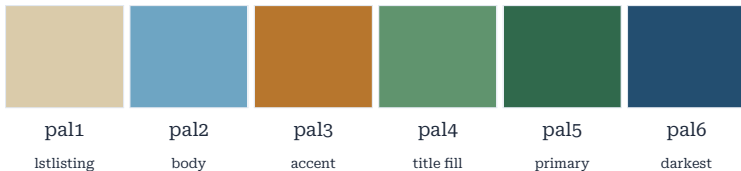
```
\OUCSetPalette{12}
```

Palette 13 – Starry Night



Starry Night — moon yellow against cypress green and night blue. `\OUCSetPalette{13}`

Palette 14 – A Thousand Li



A Thousand Li — blue-and-green landscape mineral pigments on aged silk.
`\OUCSetPalette{14}`

Palette 15 – And Quiet Flows the Don



And Quiet Flows the Don — turbid steppe browns with a cold blue and dried blood.
`\OUCSetPalette{15}`

Palette 16 – Cyberpunk Edgerunners



pal1

lstlisting

pal2

body

pal3

accent

pal4

title fill

pal5

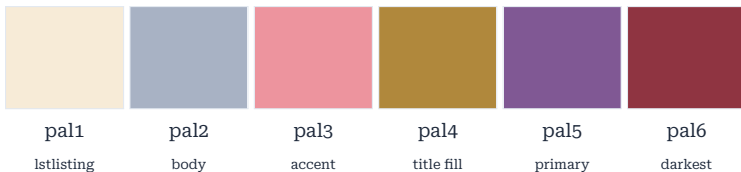
primary

pal6

darkest

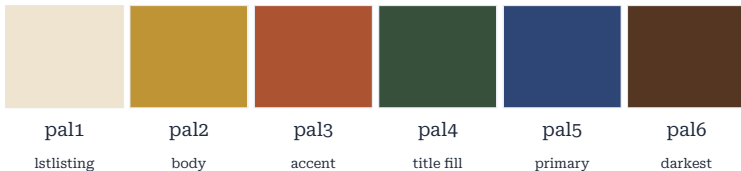
Cyberpunk Edgerunners — maximum-voltage neon over Night City black.
`\OUCSetPalette{16}`

Palette 17 – The Grand Budapest Hotel



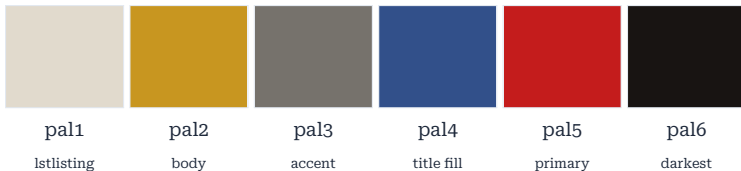
The Grand Budapest Hotel — faded aristocratic pinks and violets on a cream ground.
`\OUCSetPalette{17}`

Palette 18 – Renaissance Florence



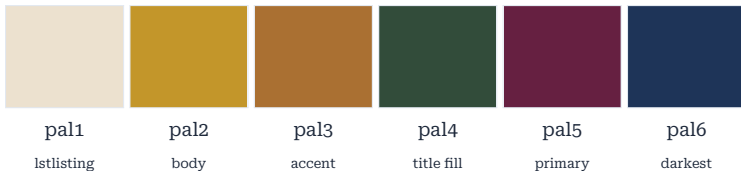
Renaissance Florence — lapis, gold leaf and terracotta in fresco proportions.
`\OUCSetPalette{18}`

Palette 19 – Soviet Avant-Garde



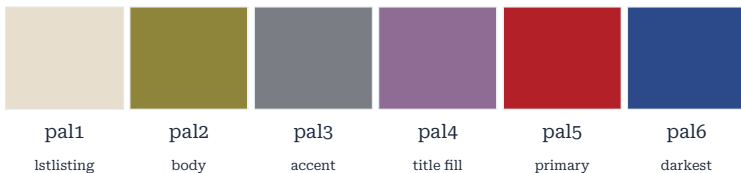
Soviet Avant-Garde — constructivist red and black with industrial greys.
`\OUCSetPalette{19}`

Palette 20 – Constantinople



Constantinople — Byzantine gilt, porphyry and the deep sea of the capital.
`\OUCSetPalette{20}`

Palette 21 – France



France — the tricolour reread through muted archival tones.

`\OUCSetPalette{21}`

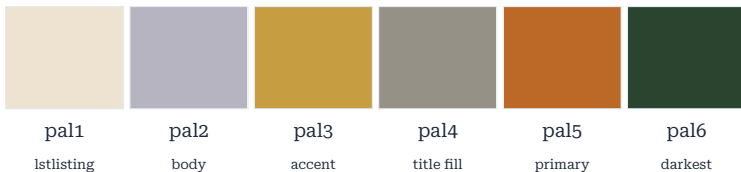
Palette 22 – Kyoto



Kyoto — zen stillness and *yūgen*; carries the Shu-hi spotcolor.

`\OUCSetPalette{22}`

Palette 23 – Siamese Dream



Siamese Dream — overexposed haze, slow-burn orange and organic dark green.
`\OUCSetPalette{23}`

This template builds on open-source work:

A Tufte-Style Book Template by Kevin Godby *et al.*

<https://www.overleaf.com/latex/templates/a-tufte-style-book-with-vdqi-title-and-contents-page/xqkjsxmrhmp>

The theorem environments (The, De, Exa, Rmk, ...), equation helpers (`\Eq`, `\EqL`), and the mathematical package selection in this template are inspired by and derived from that work.

We gratefully acknowledge Kevin Godby's contribution to the $\text{\LaTeX}$ community.

Template Ready

OUC Smart Beamer

Use template-manual.tex as your starting point

Haide College, Ocean University of China

Template guide for XeLaTeX workflow